PROJECT REPORT / HEALTHCARE DATA ANALYTICS

Health360

Comprehensive Healthcare Risk and Disease Analytics

A student project presenting sample healthcare indicators through interactive Tableau dashboards, age-group comparisons, lifestyle-factor views and a guided data story.

PROJECT
Health360: Clinical and Healthcare Risk Analysis

PROJECT TYPE
Individual student project

UPDATED
02 November 2026

ANALYTICS
Tableau Public + Web integration

PUBLISHED PROJECT WEBSITE

Open Health360 website >

This updated report points to the individual project website, published Tableau views and the owner's GitHub repository. The website and report are intended for educational project review.

Prepared as an individual educational data-visualization project.

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Project at a glance

A concise overview of the project scope, published components and report structure.

5,000	16	27	9	6
source records	original fields	prepared fields	worksheets	story scenes

PROJECT SUMMARY

Health360 organizes a supplied synthetic healthcare dataset into Tableau worksheets, an interactive dashboard and a six-scene story. The website provides a single point of access to the published analytical views and supporting project materials.

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DATA SCOPE NOTE
The 5,000 rows are records in a supplied synthetic dataset. They are not a verified count of real patients. Health360 is an educational analytics presentation, not a clinical diagnostic or treatment tool.

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Introduction and ideation

Why the project was developed and what it is designed to help a viewer explore.

2.1 PROJECT OVERVIEW

Health360: Comprehensive Healthcare Risk and Disease Analytics is an individual student project designed to make sample healthcare indicators easier to explore visually. It connects viewers to a Tableau dashboard and a companion story. Users can compare disease flags by age group and lifestyle category, review summary measures and inspect patterns in the supplied records.

2.2 PURPOSE

+ Present sample health indicators in a clear, interactive dashboard.
+ Summarize patient-record counts and the proportions of records carrying heart-disease, diabetes and stroke flags.
+ Explore patterns involving age groups, BMI quartiles, smoking, alcohol consumption and exercise-level codes.
+ Make project outputs straightforward to review through the website, Tableau links, repository and demonstration video.

2.3 PROBLEM STATEMENT

Healthcare-related indicators can be difficult to compare when they are scattered across separate values or tables. Health360 addresses this presentation problem by grouping available measures into KPI summaries, age-group and BMI comparisons, lifestyle charts, filters and a Tableau story for guided exploration.

2.4 EMPATHY MAP CANVAS

PERSPECTIVE	PROJECT INTERPRETATION
Says	I want to understand the dashboard quickly and compare groups easily.
Thinks	Which indicators and age groups show different values in the available records?
Does	Uses filters, reads KPI summaries, reviews charts and opens the story view.
Feels	Wants information to be understandable, consistent and easy to navigate.

2.5 BRAINSTORMING

+ Use KPI summaries for quick comparison of the three disease indicators and patient-record count.
+ Add filters for age group, gender, BMI quartile and lifestyle categories.
+ Use grouped bars for comparisons, a heatmap for BMI-quartile profiles and a scatter plot for clinical indicators.
+ Provide direct links to the published dashboard and story, plus a repository for the supporting files.

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Requirement analysis

The intended user journey, project requirements and high-level flow from source data to exploration.

3.1 CUSTOMER JOURNEY MAP

STAGE	USER ACTION	EXPECTED EXPERIENCE
Discover	Open the Health360 website.	Understand the project purpose and available views.
Explore	Choose an age group or another filter.	Focus on selected groups and compare applicable charts.
Interpret	Review KPI summaries, charts, tooltips and Story scenes.	Compare descriptive values while noting dataset limitations.
Share	Open published links or repository.	Present the project outputs to a reviewer.

3.2 SOLUTION REQUIREMENTS

+ Responsive web page for desktop and mobile-sized screens.
+ Published Tableau dashboard with KPI summaries, analytical charts and filters.
+ Six dashboard filters: age group, gender, BMI quartile, smoking label, alcohol label and exercise-level label.
+ Companion six-scene Tableau Story with captions aligned to the displayed views.
+ Public links to the website, Dashboard, Story and GitHub repository.

3.3 DATA FLOW - CONCEPTUAL VIEW



The diagram is a conceptual summary of the project workflow. It does not claim an undocumented live hospital-system connection or automated clinical pipeline.

3.4 TECHNOLOGY STACK

TECHNOLOGY	ROLE IN THIS PROJECT
HTML	Defines the website structure and content.
CSS	Provides page layout, styling and responsive presentation.
JavaScript	Loads the configured Tableau Embedding API v3 views and website links.
Tableau Public	Hosts the published Dashboard and companion Story.
GitHub / GitHub Pages	Stores project files and publishes the static website.
Flask	A Python app is included for use on a Python-capable host; GitHub Pages itself does not run Flask.

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Project design and planning

How the selected views, dashboard and publishing layers fit together.

4.1 PROBLEM-SOLUTION FIT

The project responds to the need to view the supplied healthcare-related indicators together. KPI summaries provide an at-a-glance overview, charts support comparisons across age and BMI groups, filters allow focused exploration, and the Tableau Story presents selected views in a guided sequence.

4.2 PROPOSED SOLUTION

+ A responsive Health360 web page introducing the purpose and limitations of the analysis.
+ A Tableau dashboard with four KPI summaries: Diabetes Rate, Heart Disease Rate, Patient Records and Stroke Rate.
+ Nine analytical worksheets covering disease prevalence, age distribution, BMI quartiles, lifestyle categories and clinical indicators.
+ Six interactive dashboard filters and a six-scene Tableau Story.
+ A GitHub repository containing source files, data notes, documentation and Tableau reference materials.

4.3 TABLEAU WORKSHEET REGISTER

WORKSHEET	VISUAL / QUESTION
01_Disease_Prevalence	Compare heart disease, diabetes and stroke flag rates across the records.
02_Age_Distribution	Show the number of records in each age group.
03_Disease_By_Age_Group	Compare the three disease rates across age groups.
04_BMI_Quartile_Profile	Show record distribution across dataset-specific BMI quartiles.
05_Disease_By_BMI_Quartile	Use a heatmap to compare the disease rates across BMI quartiles.
06_Smoking_Disease	Compare disease rates by the smoking label.
07_Alcohol_Disease	Compare disease rates by the alcohol label.
08_Exercise_Disease	Compare disease rates across exercise-level codes.
09_Clinical_Indicators	Show blood pressure versus cholesterol, with clinical and outcome context in tooltips.

4.4 SOLUTION ARCHITECTURE

PRESENTATION	ANALYTICS	PUBLISHING
Responsive HTML / CSS / JavaScript website	Tableau worksheets, dashboard and Story	Tableau Public, GitHub repository and GitHub Pages

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Testing and published results

Functional checks and the live project resources available for review.

5.1 PROJECT PLANNING

PHASE	WORK ITEM	OUTPUT / EVIDENCE
Define	Identify the problem, dataset scope and indicators.	Project scope and problem statement.
Prepare	Connect the CSV, validate records and create derived labels/calculations.	Tableau-ready data source and calculated fields.
Visualize	Build worksheets and arrange KPI summaries, charts and filters.	Nine worksheets and interactive dashboard.
Narrate	Arrange selected views into a six-scene Story.	Published Tableau Story.
Publish	Publish Tableau views and deploy the static website.	Tableau Public links and GitHub Pages site.
Review	Check links, filters, chart labels and responsive layout.	Functional review and demonstration evidence.

5.2 FUNCTIONAL AND PERFORMANCE TESTING

TEST AREA	CHECK	EXPECTED RESULT
Website loading	Open the published page in a browser.	Page content and styling appear.
Responsive layout	Check desktop and narrow-width displays.	Content remains readable and navigable.
Dashboard filters	Change age, gender, BMI or lifestyle filter.	Applicable views respond to selections.
Story navigation	Navigate all six story points.	Each scene opens with its corresponding view.
External links	Open website, Dashboard, Story and repository links.	The matching published resource opens.

Performance evidence: Actual cold-load, repeat-load and stress-test timings have not been measured in this report. The table above records functional test cases and expected outcomes, not benchmark results.

5.3 PUBLISHED OUTPUTS

RESOURCE	OPEN RESOURCE	DESCRIPTION
Health360 website	**Open website**	Published project website hosted through GitHub Pages.
Tableau Dashboard	**Open dashboard**	Health360 Clinical Healthcare Risk Analysis dashboard.
Tableau Story	**Open story**	Patient Risk and Lifestyle Insights story.
GitHub repository	**Open repository**	Source code, dataset, documentation and reference files.

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Conclusion and future scope

Project outcomes, practical benefits and the limitations that should guide interpretation.

6.1 ADVANTAGES

+ Visual summaries make the supplied indicators easier to scan and compare.
+ Six filters support focused exploration across demographic and lifestyle groups.
+ Nine worksheets, a dashboard and a Story provide multiple ways to examine the same dataset.
+ The website and repository make the project outputs easier to access and review.

6.2 LIMITATIONS

+ The dataset is described by the project owner as synthetic; it is not verified clinical evidence or a real-patient registry.
+ The source CSV contains 16 original columns. Some reference material mentions 17 fields; the actual CSV header contains 16.
+ Units and operational definitions for BMI, blood pressure, cholesterol and glucose are not documented in the supplied source.
+ Exercise codes 0, 1 and 2 have no supplied meaning. Some binary labels are inferred from field names.
+ The health_risk_score field is constant at 100.0 and is not useful as a differentiating KPI.
+ Disease flags and group differences are descriptive; they do not establish causation or individual risk.

6.3 CONCLUSION

Health360 demonstrates how a dataset can be organized into Tableau visualizations and presented through an accessible website. KPI summaries, age-group and BMI comparisons, lifestyle charts, filters and a guided Story provide several ways to explore the supplied records. The website, Tableau resources and repository provide access points for project review. The work should be treated as an educational analytics presentation, not as a medical diagnosis or treatment tool.

6.4 FUTURE SCOPE

+ Use a documented, ethically sourced dataset with clear provenance, licence and field definitions.
+ Confirm measurement units and categorical code meanings before adding clinical interpretation.
+ Add automated checks for broken links, filter behavior and data refresh consistency.
+ Conduct formal usability and performance tests across browsers and device sizes.
+ Improve accessibility through keyboard navigation, contrast checks and descriptive chart text.
+ Require privacy safeguards and appropriate clinical validation before any real-world healthcare use.

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Appendix and project resources

Clickable project links and supporting material for reviewers.

7.1 PROJECT LINKS

The links in this report open the current individual project website and the published Tableau and GitHub resources. Select the blue link text to open each destination.

RESOURCE	CLICKABLE LINK	PURPOSE
Health360 website	**Open Health360 website**	Main project landing page and entry point to the published views.
Tableau Dashboard	**Open Tableau Dashboard**	Interactive KPI summaries, charts and filters.
Tableau Story	**Open Tableau Story**	Six-scene guided presentation of selected views.
GitHub repository	**Open GitHub repository**	Website source, Flask app, dataset, documentation and Tableau files.
Source dataset	**Open CSV file**	Supplied synthetic healthcare dataset, 5,000 records and 16 source fields.

7.2 SOURCE FILES

+ Dataset: smart_healthcare_dataset.csv (5,000 records, 16 original columns).
+ Tableau workbook: published as Health360_Clinical_Healthcare_Risk_Analysis on Tableau Public.
+ Tableau reference material: Business Question and Visualisation Report, Dashboard Design Template and Story Design Template are stored in the Tableau folder of the GitHub repository.
+ Website source: index.html and the assets folder; a Flask app and deployment configuration are also included for Python-capable hosting.

7.3 DEMONSTRATION VIDEO

Demo video link
To be added after the final screen-recorded demonstration is recorded and uploaded. The previous Google Drive link has been removed because it does not represent the new individual project video.

7.4 DATA AND INTERPRETATION NOTE

The project owner describes the dataset as synthetic. The source publisher or licence was not documented in the supplied material. Several units and code definitions require confirmation. The report therefore describes the dashboard as an exploratory visualization project and does not present its values as clinical guidance.

END OF REPORT | HEALTH360 - CLINICAL AND HEALTHCARE RISK ANALYSIS